Fri, Oct 2 at 7:54 PM

You said:




Here is my solution to the problem, I first asked if there are n observations, how many numbers of observations will be fallen into a,b. Then this number divided by n is our theta hat

07:54 PM

ChatGPT said:

Your main idea is correct. The clean way to formalize it is to turn “falls between a and b” into an indicator variable.

Since

$$\hat{\theta} = \hat{F}_n(b) - \hat{F}_n(a),$$

and

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} I(X_i \le x),$$

we get

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} \left[I(X_i \le b) - I(X_i \le a) \right].$$

Ignoring the endpoint issue for a continuous F, this is

$$\boxed{\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} I(a < X_i \le b)}$$

So exactly as you said:

> count how many of the n observations fall in $(a, b]$, then divide by n.

Now define

$$Y_i = I(a < X_i \leq b).$$

For each observation,

$$P(Y_i = 1) = P(a < X_i \leq b) = F(b) - F(a) = \theta.$$

Therefore

$$Y_i \sim \text{Bernoulli}(\theta).$$

Hence the total number falling in the interval,

$$K = \sum_{i=1}^{n} Y_i,$$

satisfies

$$\boxed{K \sim \text{Binomial}(n, \theta)}.$$

And

$$\hat{\theta} = \frac{K}{n}.$$

This is where one small correction to your handwritten solution matters. You correctly wrote

$$\text{Var}(K) = n\theta(1 - \theta),$$

but for $\hat{\theta} = K/n$,

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{K}{n}\right) = \frac{1}{n^2}\,\text{Var}(K).$$

Therefore

$$\boxed{\text{Var}(\hat{\theta}) = \frac{\theta(1 - \theta)}{n}}.$$

So the true standard error is

$$\boxed{SE(\hat{\theta}) = \sqrt{\frac{\theta(1-\theta)}{n}}}.$$

But θ is unknown, so we estimate it using $\hat{\theta}$:

$$\boxed{\widehat{SE}(\hat{\theta}) = \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}}.$$

That is the answer to the first part.

For the confidence interval, for large n, the CLT gives

$$\frac{\hat{\theta} - \theta}{\sqrt{\theta(1-\theta)/n}} \approx N(0,1).$$

Replacing the unknown θ in the standard error by $\hat{\theta}$,

$$\boxed{\theta \approx \hat{\theta} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}}$$

is an approximate $1-\alpha$ confidence interval.

For example, for a 95% CI,

$$\boxed{\hat{\theta} \pm 1.96\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}}.$$

So your reasoning can be summarized very neatly as

$$\boxed{\#\{X_i \in (a,b]\} \sim \text{Binomial}(n,\theta)}$$

then

$$\boxed{\hat{\theta} = \frac{\#\{X_i \in (a,b]\}}{n}}$$

and therefore

$$\boxed{E[\hat{\theta}] = \theta, \qquad \mathrm{Var}(\hat{\theta}) = \frac{\theta(1-\theta)}{n}.}$$

The conceptual point you had is exactly the right one: **this empirical-CDF problem is secretly just estimating a binomial proportion.**

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You said:

summarize my ideas of creating a K (which is binomial) and then our interested is K/n. without using indicator, give me a paragraph for my argument

07:57 PM

ChatGPT said:

Let K be the number of observations among $X_1, \ldots, X_n$ that fall in the interval $(a, b]$. Since each observation has probability

$$\theta = P(a < X \leq b) = F(b) - F(a)$$

of falling in this interval, and the observations are independent, we have

$$K \sim \mathrm{Binomial}(n, \theta).$$

Our estimator $\hat{\theta}$ is the proportion of observations that fall in $(a, b]$, so

$$\hat{\theta} = \frac{K}{n}.$$

Therefore, using the mean and variance of a binomial random variable,

$$E(\hat{\theta}) = \frac{E(K)}{n} = \theta,$$

and

$$\mathrm{Var}(\hat{\theta}) = \frac{\mathrm{Var}(K)}{n^2} = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n}.$$

Thus the estimated standard error is

$$\widehat{SE}(\hat{\theta}) = \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}.$$

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You said:

is my thinking correct?

08:01 PM

ChatGPT said:

Yes. Your thinking is correct.

You are recognizing that the quantity

$$\theta = F(b) - F(a)$$

is just the probability that one observation falls in $(a, b]$. So among n independent observations, if K counts how many fall in that interval, then

$$K \sim \text{Binomial}(n, \theta).$$

Then the empirical proportion of observations in the interval is

$$\hat{\theta} = \frac{K}{n}.$$

That is exactly the right way to think about the problem.

The only thing to be careful about is the variance scaling. Since K has variance $n\theta(1-\theta)$,

$$\text{Var}\left(\frac{K}{n}\right) = \frac{1}{n^2}\,\text{Var}(K) = \frac{\theta(1-\theta)}{n}.$$

So your conceptual setup is fully correct; the main technical step is remembering that dividing a random variable by n divides its variance by n^2.

08:01 PM

Fri, Oct 2 at 9:04 PM

You said:

what is the process of selecting bootstrap sample from eCDF

09:04 PM

ChatGPT said:

Suppose your original sample is

$$X_1, \ldots, X_n.$$

The empirical CDF $\hat{F}_n$ puts probability $1/n$ on each observed data point. So drawing a bootstrap sample "from the eCDF" simply means sampling from your observed values **with replacement**.

For example, if your original data are

$$(2, 5, 7, 10),$$

then the empirical distribution is

$$P^*(X = 2) = P^*(X = 5) = P^*(X = 7) = P^*(X = 10) = \frac{1}{4}.$$

A bootstrap sample of size 4 might be

$$(5, 5, 10, 2).$$

Another could be

$$(7, 2, 7, 7).$$

Duplicates are allowed because you are sampling with replacement.

Conceptually, the process is

$$X_1, \ldots, X_n \quad \longrightarrow \quad \hat{F}_n \quad \longrightarrow \quad X_1^*, \ldots, X_n^* \overset{iid}{\sim} \hat{F}_n.$$

Then for each bootstrap sample, you recalculate your statistic. For your earlier problem,

$$\hat{\theta} = \hat{F}_n(b) - \hat{F}_n(a),$$

so for bootstrap sample b,

$$\hat{\theta}^* = \hat{F}_n^*(b) - \hat{F}_n^*(a).$$

You repeat this many times, say $B = 1000$ or 10000, producing

$$\hat{\theta}_1^*, \hat{\theta}_2^*, \ldots, \hat{\theta}_B^*.$$

The spread of these bootstrap estimates approximates the sampling uncertainty of $\hat{\theta}$.

The key intuition is: **we do not know the true F, so we use the empirical distribution $\hat{F}_n$ as a stand-in for F, and repeatedly pretend $\hat{F}_n$ is the population.**

09:04 PM

You said:

how do we draw the bootstrap sample, based on what probability

09:04 PM

ChatGPT said:

From the empirical distribution itself.

If your observed sample is

$$X_1, \ldots, X_n,$$

then the empirical distribution $\hat{F}_n$ assigns probability

$$\boxed{\frac{1}{n}}$$

to each observed data point.

So in a bootstrap draw,

$$P(X^* = X_i) = \frac{1}{n}, \qquad i = 1, \ldots, n.$$

For example, if the original sample is

$$(2, 5, 7, 10),$$

then the bootstrap distribution is

$$P(X^* = 2) = P(X^* = 5) = P(X^* = 7) = P(X^* = 10) = \frac{1}{4}.$$

Each bootstrap observation is drawn independently from those four values with those probabilities.

So to form one bootstrap sample of size $n = 4$, you draw four times:

$$X_1^*, X_2^*, X_3^*, X_4^* \overset{iid}{\sim} \hat{F}_n.$$

You might get

$$(5, 2, 5, 10).$$

The important idea is that the probabilities do not come from the unknown true F. We replace F by $\hat{F}_n$, and $\hat{F}_n$ says, empirically, that each observed data point has mass $1/n$.

One subtle point: if some observed values are repeated, then a value can have probability larger than $1/n$. For example, if your data are

$$(2, 2, 5, 7),$$

then

$$P_{\hat{F}_n}(X^* = 2) = \frac{2}{4} = \frac{1}{2},$$

because 2 appeared twice in the original sample.

09:05 PM

You said:



Don't give me the answer yet
I am stuck at the later 2 questions. I am struggling to find out E of Fn (not hat) of (X*n bar). give me some hints

09:31 PM

ChatGPT said:

For the last two quantities, the main difficulty is that you are no longer conditioning on the observed sample.

Earlier, when you compute

$$E_{\hat{F}_n}(\bar{X}_n^*) = E(\bar{X}_n^* \mid X_1, \ldots, X_n),$$

you treat $X_1, \ldots, X_n$ as fixed numbers.

But for

$$E_F(\bar{X}_n^*)$$

you now have to remember that the original observations $X_1, \ldots, X_n$ are themselves random under F.

The key hint is to use the law of total expectation:

$$E_F(\bar{X}_n^*) = E_F\left[E(\bar{X}_n^* \mid X_1, \ldots, X_n)\right].$$

You already know, or can derive, the inner conditional expectation from the earlier part. Then take expectation of that quantity under the original sampling distribution F.

For the variance, use the law of total variance:

$$\mathrm{Var}_F(\bar{X}_n^*) = E_F\left[\mathrm{Var}(\bar{X}_n^* \mid X_1, \ldots, X_n)\right] + \mathrm{Var}_F\left[E(\bar{X}_n^* \mid X_1, \ldots, X_n)\right].$$

So the structure is:

$$\boxed{\text{unconditional mean} = E(\text{conditional mean})}$$

and

$$\boxed{\text{unconditional variance} = E(\text{conditional variance}) + \mathrm{Var}(\text{conditional mean})}$$

For the variance part, you will eventually need to simplify something involving the empirical variance of the original sample. A useful intermediate target is to figure out

$$E_F\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\right].$$

That quantity is slightly smaller than the population variance because the denominator is n, not $n-1$.

If you want, try the expectation part first and send me your next line; I can tell you whether you're on the right track without giving away the result.

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